

Rimmel Demand Forecasting System

Dataset Architecture, Data Lineage & Production Model Governance Guide

Executive Purpose: This document provides executive and business stakeholders with complete transparency into the data source, temporal boundaries, training methodology, and forecasting logic used by the certified **Exp6 Production Forecasting Model**. It establishes verified mathematical lineage from the raw transaction file to the 10-day forward demand projections (September 11–20, 2026).

1. Core Timeline & Operational Horizons

Operational Phase	Exact Calendar Period	Duration	Observations	Business Purpose
Historical Baseline	01 Aug 2025 → 31 Aug 2026	396 Days	559,548	Baseline demand pattern learning across all 4 platforms.
Holdout Validation	01 Sep 2026 → 10 Sep 2026	10 Days	14,130	Strictly unseen evaluation. Verified +2.52% net bias.
Final Production Training	01 Aug 2025 → 10 Sep 2026	406 Days	573,678	Refit on full 406 days to capture freshest September run-rates.
Forward Forecast	11 Sep 2026 → 20 Sep 2026	10 Days	14,130 (proj.)	Active operational replenishment forecast (1,934 units).

2. Data Source & Physical Inventory Governance

Primary Raw Data File: Rimmel Brand Sales Data - 1 Jan 2025 to 10 Sep 2026.xlsx (7.84 MB, 101,085 raw sales transactions). An immutable SHA-256 byte-level backup is preserved at data/raw/ORIGINAL_COPY.xlsx.

Catalog Scope: Exactly **674 unique canonical SKUs** mapped across **4 commercial platforms**: Amazon, eBay, Website, and Other (B2B / MFN manual orders). This defines exactly **1,413 active SKU × Platform selling series** across the enterprise catalog.

Shared Warehouse Stock Rule (Crucial Inventory Policy): Physical inventory is held in **ONE central warehouse pool**. Amazon, eBay, Website, and Other draw from the same stock. **Inventory is NEVER summed across platforms**. All Days of Cover (DoC) and replenishment decisions divide central shared warehouse stock by total physical catalog demand.

3. Dataset Composition & Metrics Summary

Verified Metric	Value	Operational Significance
Total Daily Observations	573,678	406 days × 1,413 SKU-platform combinations (Zero-filled grid)
Total Physical Units Conserved	164,786.0 units	100% mathematical conservation from raw Excel transactions
Canonical Catalog SKUs	674 SKUs	Cleaned, verified master catalog with family & category mapping
Sales Platforms	4 Platforms	Amazon (retail), eBay, Direct Website, Other (B2B/MFN)
Intermittent Zero Days	> 75% of series-days	Handled via ZERO treatment to eliminate phantom demand

4. Information & Features Used by the Model

The model does not rely on simple trend extrapolation. It utilizes **74 causal, strictly past-dated features** rolling backward from \$T-1\$:

- **Historical Sales Velocity:** Rolling averages over 7, 14, 30, 90, 180, and 365 calendar days (\$v_7\$ through \$v_{365}\$).
- **Demand Lags:** Sales volumes observed exactly 1, 7, 14, 30, 90, 180, and 365 days prior.
- **Demand Stability & Volatility:** 30-day and 90-day Coefficient of Variation (\$CV_{30}\$, \$CV_{90}\$) measuring intermittency.
- **Shared Warehouse Inventory:** Live stock level, in-stock boolean indicator, stockout flag, and days since last stockout.
- **Platform & Channel Signals:** Amazon session traffic momentum, buy-box ownership percentage, and eBay promotional flags.
- **Calendar & Seasonality:** Day of week, weekend indicator, month of year, and day-in-month seasonality patterns.

5. Two-Stage MLOps Training & Calibration Protocol

To guarantee mathematical rigor and business reliability, the system followed an airtight 2-stage lifecycle:

Stage 1 — Retrospective Holdout Validation (Sep 1–10, 2026): The model was first trained strictly on data prior to Sep 1, 2026. It was then tested on the unseen 10-day holdout. The model predicted **2,121.2 units** against **2,069.0 actual units**, delivering an extraordinary **+2.52% net portfolio bias**.

Stage 2 — Full Production Training Refit (Aug 1, 2025–Sep 10, 2026): Once the Exp6 architecture was certified, the model was refitted on all 406 days of available history. This ensured the forward forecast leverages the freshest early-September sales velocity and latest warehouse stock levels.

Stage 3 — Combined Post-Hoc Calibration: To protect against beauty e-commerce sparsity and active stockouts, Exp6 applies two frozen business rules: (1) *Zero-demand suppression* (\$\alpha=0.10\$) when a SKU has zero sales over 30 days and no marketing push; (2) *Stockout dampening* (\$\eta=0.10\$) when warehouse stock is zero, preventing unfulfillable order projections.

6. End-to-End Data Lineage Architecture

Step	Layer / File	Input Data	Output Transformation
1. Raw Ingestion	src/data_ingestion.py	Client Excel (101k rows)	Attaches immutable source_row_id; creates raw backup.
2. Relational DB	data/rimmel_clean.db	Normalized sales	Stores raw_transactions & builds 406-day Cartesian grid.
3. Feature Layer	ml_features_zero table	Cartesian grid (573k rows)	Computes 74 causal lag, rolling, stock, and promo features.
4. Model Refit	models/production_lgbm_model.pkl	All 406 days (573,678 rows)	Fits 150 LightGBM decision trees (max_depth=6, lr=0.05).
5. Client Reports	reports/*.xlsx & data/processed/*.csv	Sep 10 feature snapshot	Generates 10-day SKU summary workbooks & fast dashboard caches.

Model Certification & Data Integrity Status: PASSED (100% VERIFIED)

- Zero data leakage between training and validation windows.
- 100% mathematical conservation of raw sales units.
- Certified for procurement, replenishment, and executive multi-channel demand planning.